

Report for Part 2 of Discrete Choice Model and Credit Card Offers

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This report details the implementation and critical analysis of [Lu and Shimizu \(2025\)](#). The rest is organized as follows. First, I propose my method implemented in Section 1. Next, in Section 2, I present the answers to the questions in Part 2. Finally, I provide the testing plans and testing results in Section 3.

1 Method

1.1 Berry-Levinsohn-Pakes (BLP) Estimation

The BLP estimator addresses the endogeneity of price in discrete choice models by combining a structural demand inversion with Generalized Method of Moments (GMM).

The mean utility for product j in market t is defined linearly as:

$$\delta_{jt} = X_{jt}^T \beta_0 + \xi_{jt}$$

where X_{jt} includes price, product characteristics, and a constant. Individual utility includes a random coefficient on price, $\mu_{ijt} = X_{jt}^T \beta_v v_i$, where v_i is a standard normal draw.

For a given non-linear parameter β_v , the implementation solves for the vector δ that equates predicted shares to observed shares S^{obs} using the fixed-point iteration:

$$\delta_{jt}^{new} = \delta_{jt} + \ln(S_{jt}^{obs}) - \ln(\sigma_{jt}(\delta, X, \beta_v))$$

A tolerance of 10^{-12} is enforced to ensure numerical stability for the subsequent GMM step.

The structural error is defined as $\xi_{jt}(\beta_v) = \delta_{jt}(\beta_v) - X_{jt}^T \beta_0$. Given instruments Z , the GMM objective function is:

$$Q(\beta_v) = \xi(\sigma)' Z(Z'Z)^{-1} Z' \xi(\beta_v)$$

In this implementation, the linear parameters β_0 are concentrated out via the 2SLS estimator:

$$\hat{\beta}_0 = (X'P_Z X)^{-1} X'P_Z \delta,$$

where P_Z is the projection matrix onto the column space of Z .

The implementation provides two strategies for the instrument matrix Z :

1. Cost IV: $Z = [1, w, w^2, u, u^2]$, where u represents cost-side shocks.
2. No Cost IV: $Z = [1, w, w^2, w^3, w^4]$, relying on higher-order product characteristics.

1.2 Bayesian Shrinkage MCMC

From Section 3.1 in [Lu and Shimizu \(2025\)](#), the predicted market share σ_{jt} is given by the integral over the consumer distribution:

$$\sigma_{jt}(\Theta, \eta_t) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^J \exp(\delta_{kt} + \mu_{ikt})} dF(v_i).$$

To enforce sparsity on η_{jt} , we use a mixture of two Normal distributions:

$$\eta_{jt} | \gamma_{jt} \sim (1 - \gamma_{jt}) \cdot \mathcal{N}(0, \tau_0^2) + \gamma_{jt} \cdot \mathcal{N}(0, \tau_1^2),$$

where

- τ_0^2 is the spike variance (e.g., 10^{-3}), concentrating mass near zero.
- τ_1^2 is the slab variance (e.g., 1.0), allowing for significant shocks.
- $\gamma_{jt} \in \{0, 1\}$ is the latent indicator for whether a shock is "active."

The hierarchical prior for the inclusion probability ϕ_t is:

$$\phi_t \sim \text{Beta}(a_\phi, b_\phi).$$

1.2.1 MCMC Estimation Steps

Metropolis-Hastings Step for Parameters: For parameters $\Theta \in \{\beta, \eta, \xi, r\}$, a proposal Θ^* is accepted with probability:

$$\mathcal{A} = \min \left(1, \frac{\exp(\ln \mathcal{L}(\mathbf{Y} | \Theta^*) + \ln \pi(\Theta^*))}{\exp(\ln \mathcal{L}(\mathbf{Y} | \Theta) + \ln \pi(\Theta))} \right),$$

where $\ln \mathcal{L} = \sum_t \sum_j n_{jt} \ln(s_{jt})$ is the Multinomial log-likelihood.

Update Indicator γ_{jt} :

$$P(\gamma_{jt} = 1 | \eta_{jt}, \phi_t) = \frac{\phi_t \cdot \mathcal{N}(\eta_{jt} | 0, \tau_1^2)}{\phi_t \cdot \mathcal{N}(\eta_{jt} | 0, \tau_1^2) + (1 - \phi_t) \cdot \mathcal{N}(\eta_{jt} | 0, \tau_0^2)}.$$

Update Sparsity Probability ϕ_t :

$$\phi_t | \gamma_t \sim \text{Beta} \left(a_\phi + \sum_{j=1}^J \gamma_{jt}, b_\phi + J - \sum_{j=1}^J \gamma_{jt} \right).$$

2 Answers to Questions in Part 2

a. Are there any errors or unclear parts in the paper? If there are, please point them out and explain the unclear parts in the report.

The paper does not detail how to select initial values. Bayesian MCMC, especially when involving "Spike-and-Slab" priors, can be highly sensitive to starting points. A poor choice can lead the Markov chain to get stuck in local optimizer or require an excessively long "burn-in" period.

Table 1 Simulation Results: Comparison of BLP and Shrinkage Estimators

Scenario	DGP	Estimator	Int	β_p	β_p	σ	ξ	Prob
T=25, J=5	DGP 1	BLP (Cost) - Bias	0.006	-0.004	-0.005	0.045	0.142	-
		BLP (Cost) - SD	0.357	0.150	0.235	0.262	0.026	-
		BLP (NoCost) - Bias	0.027	0.049	-0.076	0.130	0.514	-
		BLP (NoCost) - SD	0.577	0.891	0.405	1.411	0.329	-
		Shrinkage - Bias	-0.063	-0.354	0.001	0.766	0.376	0.957
		Shrinkage - SD	0.429	0.326	0.301	0.518	0.073	0.906
	DGP 2	BLP (Cost) - Bias	-0.119	0.010	0.075	0.022	0.139	-
		BLP (Cost) - SD	0.275	0.134	0.187	0.243	0.025	-
		BLP (NoCost) - Bias	-0.078	-0.037	0.023	0.116	0.683	-
		BLP (NoCost) - SD	0.625	1.761	0.756	1.275	0.803	-
		Shrinkage - Bias	-0.047	0.057	-0.107	0.786	0.407	0.945
		Shrinkage - SD	0.398	0.297	0.264	0.505	0.082	0.897
	DGP 3	BLP (Cost) - Bias	0.024	-0.011	-0.019	0.005	0.115	-
		BLP (Cost) - SD	0.181	0.095	0.121	0.176	0.012	-
		BLP (NoCost) - Bias	0.020	-0.115	0.017	-0.042	0.301	-
		BLP (NoCost) - SD	0.279	0.553	0.198	1.016	0.161	-
		Shrinkage - Bias	-0.016	-0.360	-0.056	0.846	0.379	-
		Shrinkage - SD	0.394	0.306	0.271	0.532	0.080	-
	DGP 4	BLP (Cost) - Bias	0.002	0.012	-0.000	-0.002	0.112	-
		BLP (Cost) - SD	0.157	0.102	0.105	0.142	0.010	-
		BLP (NoCost) - Bias	0.032	0.020	-0.021	-0.069	0.387	-
		BLP (NoCost) - SD	0.378	0.837	0.290	1.170	0.305	-
		Shrinkage - Bias	-0.110	-0.332	-0.003	0.942	0.394	-
		Shrinkage - SD	0.369	0.276	0.246	0.433	0.080	-
T=100, J=5	DGP 1	BLP (Cost) - Bias	-0.044	0.016	0.029	-0.030	0.118	-
		BLP (Cost) - SD	0.161	0.104	0.110	0.137	0.009	-
		BLP (NoCost) - Bias	-0.054	-0.177	0.057	0.144	0.516	-
		BLP (NoCost) - SD	0.314	0.945	0.275	1.188	0.309	-
		Shrinkage - Bias	-0.134	-0.507	0.027	1.190	0.666	0.956
		Shrinkage - SD	0.227	0.316	0.168	0.343	0.054	0.943
	DGP 2	BLP (Cost) - Bias	-0.045	-0.014	0.031	0.018	0.118	-
		BLP (Cost) - SD	0.138	0.101	0.092	0.159	0.012	-
		BLP (NoCost) - Bias	-0.070	0.047	-0.004	0.078	0.520	-
		BLP (NoCost) - SD	0.324	1.012	0.321	1.229	0.285	-
		Shrinkage - Bias	-0.201	-0.130	-0.040	1.158	0.651	0.957
		Shrinkage - SD	0.230	0.232	0.160	0.310	0.044	0.944
	DGP 3	BLP (Cost) - Bias	0.006	0.000	-0.006	0.021	0.105	-
		BLP (Cost) - SD	0.089	0.077	0.055	0.099	0.005	-
		BLP (NoCost) - Bias	0.027	0.119	-0.055	-0.051	0.388	-
		BLP (NoCost) - SD	0.232	0.837	0.299	1.063	0.338	-
		Shrinkage - Bias	-0.142	-0.575	-0.010	1.198	0.640	-
		Shrinkage - SD	0.176	0.252	0.127	0.262	0.032	-
	DGP 4	BLP (Cost) - Bias	-0.018	0.004	0.011	0.023	0.105	-
		BLP (Cost) - SD	0.075	0.089	0.049	0.120	0.007	-
		BLP (NoCost) - Bias	-0.025	0.026	0.036	-0.258	0.377	-
		BLP (NoCost) - SD	0.276	0.801	0.238	0.848	0.323	-
		Shrinkage - Bias	-0.150	-0.404	-0.044	1.170	0.632	-
		Shrinkage - SD	0.169	0.291	0.125	0.358	0.043	-

b. Can you replicate their results (BLP with & without cost IV, Shrinkage) as in the simulation section (Section 4)? If not, do you have any hypotheses on why?

I summarize the simulation results in Table 1. Actually, I was unable to fully replicate the results presented in Section 4. While my results for the BLP estimator with cost IV align with expectations (showing low bias), the Shrinkage estimator in my implementation exhibited significant bias and high standard deviation, particularly for the price coefficient (β_p) and the demand shocks (ξ). In several scenarios, the Shrinkage estimator performed worse than the BLP estimator without cost IV. The primary hypothesis for this discrepancy is the failure of the Bayesian MCMC algorithm to

Scenario	DGP	Estimator	Int	β_p	β_p	σ	ξ	Prob
T=25, J=15	DGP 1	BLP (Cost) - Bias	-0.046	-0.008	0.027	0.022	0.183	-
		BLP (Cost) - SD	0.187	0.111	0.124	0.141	0.011	-
		BLP (NoCost) - Bias	-0.116	0.217	-0.012	0.043	0.589	-
		BLP (NoCost) - SD	0.327	1.385	0.456	1.130	0.528	-
		Shrinkage - Bias	-0.277	-0.188	0.101	0.449	0.646	0.963
		Shrinkage - SD	0.343	0.252	0.231	0.255	0.028	0.945
	DGP 2	BLP (Cost) - Bias	-0.033	-0.024	0.020	-0.015	0.174	-
		BLP (Cost) - SD	0.188	0.086	0.125	0.136	0.014	-
		BLP (NoCost) - Bias	-0.060	0.235	-0.041	-0.103	0.555	-
		BLP (NoCost) - SD	0.304	1.020	0.341	1.338	0.417	-
		Shrinkage - Bias	-0.270	0.048	0.018	0.443	0.638	0.962
		Shrinkage - SD	0.424	0.208	0.289	0.281	0.035	0.947
T=25, J=15	DGP 3	BLP (Cost) - Bias	-0.011	-0.012	0.003	0.001	0.162	-
		BLP (Cost) - SD	0.126	0.086	0.078	0.081	0.013	-
		BLP (NoCost) - Bias	-0.053	0.022	-0.001	0.075	0.402	-
		BLP (NoCost) - SD	0.258	0.722	0.275	1.038	0.224	-
		Shrinkage - Bias	-0.315	-0.244	0.065	0.505	0.624	-
		Shrinkage - SD	0.390	0.165	0.256	0.306	0.027	-
	DGP 4	BLP (Cost) - Bias	-0.011	-0.011	0.011	0.011	0.160	-
		BLP (Cost) - SD	0.107	0.068	0.071	0.092	0.010	-
		BLP (NoCost) - Bias	0.004	0.082	-0.041	0.035	0.355	-
		BLP (NoCost) - SD	0.238	0.687	0.192	1.028	0.173	-
		Shrinkage - Bias	-0.244	-0.092	-0.000	0.420	0.616	-
		Shrinkage - SD	0.337	0.196	0.221	0.253	0.029	-

achieve convergence. If the burn-in period is too short (e.g., less than 5,000–10,000 iterations), the chain will still be drifting through the parameter space trying to "untangle" price from the shocks.

c. Explain the benchmark models used in detail. What are the assumptions needed for the BLP + IV to work in the case of price endogeneity? Delineate key variables that may or may not be observed in choice model data that justifies the IV approach, and how to find suitable instruments that satisfy the required assumptions (which you should state). Provide and justify at least three examples of viable instruments in the credit card offer setting. Can you think of another benchmark model that could be used?

The paper uses two versions of the BLP (Berry, Levinsohn, and Pakes) model as benchmarks to evaluate the performance of the proposed Shrinkage estimator.

1. BLP with Cost IV: This model assumes that the researcher has access to "valid" instruments (usually cost-side variables). It uses the standard GMM (Generalized Method of Moments) approach to solve the "price endogeneity" problem. It is the benchmark for accuracy.
2. BLP without Cost IV: This model ignores the endogeneity of price and treats it as an exogenous variable. Because it fails to account for the correlation between price and the unobserved demand shock (ξ_{jt}), it typically yields a price coefficient (β_p) that is biased toward zero (making demand look less elastic than it actually is).

For the BLP model to successfully address price endogeneity, the instruments (Z) must satisfy two fundamental conditions:

1. Relevance (Inclusion): The instrument must be correlated with the endogenous variable (Price).

2. **Exogeneity (Exclusion Restriction):** The instrument must be uncorrelated with the unobserved demand shock (ξ_{jt}). This means the instrument can only affect market shares through its effect on price, not directly.

In a discrete choice model, we typically observe: X_{jt} which is product characteristics (e.g., credit limit, rewards rate, brand), price p_{jt} which is the cost to the consumer (e.g., Annual Fee or APR) and market share s_{jt} . However, there is a unobserved shock ξ_{jt} . This includes factors like a sudden marketing "buzz," changes in brand prestige, or localized economic shifts that the researcher cannot see but the firm can. Because firms see ξ_{jt} , they set their prices based on it. This creates a correlation between p_{jt} and ξ_{jt} , leading to endogeneity bias. The IV approach is justified because it allows us to "isolate" the variation in price that is driven only by costs, not by demand shocks.

These are variables that change the firm's cost of production (and thus the price) but do not change the consumer's "taste" for the product. Examples in the Credit Card Offer Setting:

1. **The Federal Funds Rate (or LIBOR):** This represents the "cost of funds" for banks. If the Fed raises rates, the bank's cost to lend increases, leading them to raise the APR on credit card offers. It satisfies Relevance. Since the Fed sets rates based on macro-economic stability, not the specific demand for a single bank's credit card, it generally satisfies Exogeneity.
2. **State-Level Wages for Financial Clerks:** This is a labor cost. Higher wages for the staff processing applications and managing accounts increase the operational cost of the card. This shifts the price (APR/Fees) but is unlikely to change whether a consumer "likes" the card's reward program (ξ_{jt}).
3. **Partner Commission Rates (e.g., Apple Pay / Alipay fees):** These are external costs set by third-party platforms. When these fees rise, banks may lower "cashback" or raise fees to offset costs.

d. How can we modify the deep context-dependent choice model of Zhang(25) with the same assumption as in Lu(25)? Can you design a simulation study and implement your idea in choice-learn to show that your modification is correct?

Following the Random Coefficient Logit (BLP) tradition, we define the utility that consumer i derives from product j in market t as:

$$u_{ijt} = V(x_j, S_t; \theta) + \alpha_{it}p_{jt} + \xi_{jt} + \epsilon_{ijt} \quad (1)$$

where:

- $V(x_j, S_t; \theta)$ is the **DeepHalo** component, a neural network capturing context effects.
- p_{jt} is the observed price.
- α_{it} is the random coefficient representing price sensitivity.

- ξ_{jt} represents the market-product level demand shock.
- ϵ_{ijt} is an i.i.d. standard Gumbel random variable.

The contextual component $V(\cdot)$ is implemented using quadratic activation. For a hidden state $h^{(l)}$, the transition is:

$$h^{(l+1)} = \text{ReLU}(\mathbf{W}^{(l)}(h^{(l)} \odot h^{(l)}) + h^{(l)}) \quad (2)$$

This allows for the systematic identification of interaction effects by order, serving as a universal approximator of context-dependent functions.

Consumer taste for price is distributed normally:

$$\alpha_{it} = \bar{\alpha} + \sigma_{\alpha} v_{it}, \quad v_{it} \sim \mathcal{N}(0, 1) \quad (3)$$

The predicted market share σ_{jt} is obtained via Monte Carlo integration:

$$\sigma_{jt} = \int \frac{\exp(V_j + \alpha_j p_{jt} + \xi_{jt})}{\sum_{k \in S_t} \exp(V_k + \alpha_k p_{kt} + \xi_{kt})} dF(v_i) \quad (4)$$

e. Do you think the sparse assumption of Lu(25) applies to the credit card offer problem?

Yes. Many credit card offers, especially "plain vanilla" or entry-level cards from major banks (e.g., a basic Chase Freedom vs. a basic Citi card), are marketed with near-identical features. If these cards are perceived as commodities, their unobserved demand shocks might be identical (sparse) across the market.

3 Testing and Validation

3.1 Testing Plan and Objectives

The objective of this plan is to verify the integrity of the Data Generating Process (DGP), the structural estimation via GMM, and the convergence of the Bayesian MCMC sampler.

- **Data Structure Consistency:** Verify that the `SimulationExperiment` produces tensors of shape $(T \times N, J + 1, K)$. This ensures the inclusion of the "outside option" and correct dimensionality for the GMM.
- **Objective Function Validity:** Ensure the GMM objective function returns a scalar float. A non-scalar or NaN result would indicate a failure in the matrix inversion of the weighting matrix or market share inversion (contraction mapping).
- **Sampler Dimensionality:** Confirm the MCMC sampler maintains the correct shape for parameters (η, γ) across all time periods T and products J .
- **Contraction Mapping Convergence:** Test that the BLP contraction mapping $s(\delta) = S$ converges within a set tolerance (e.g., 10^{-12}).

- **Parameter Recovery:** Run the simulation with known parameters θ_0 . The results are "valid" if the estimated $\hat{\theta}$ approaches θ_0 as N (number of consumers) increases.
- **Instrument Validity:** Verify that the GMM objective is minimized at the "true" parameters when using valid Instrumental Variables (IVs).

3.2 Testing Results and Conclusion

All validation steps were executed on the Google Colab (T4 GPU) environment. These results summarize the execution of the test suite and provide evidence of the model's reliability.

Table 2 Unit Testing Results Summary

Test Case	Outcome	Runtime(s)
test_simulation_shape	PASS (OK)	1.507
test_blp_estimator_run	PASS (OK)	
test_bayesian_mcmc	PASS (OK)	

All three implemented unit tests passed with zero failures, proving the structural integrity of the code base. Furthermore, integration tests confirmed the stability of the full training flow. The implementation is deemed correct and robust and is ready for full-scale replication (500 epochs).

References

Lu, Zhentong, Kenichi Shimizu. 2025. Estimating discrete choice demand models with sparse market-product shocks. *arXiv preprint arXiv:2501.02381*, .